Rouché's Theorem

An End-to-End Meta-Math Walkthrough

Dibakar Sigdel[1]

[1] *Mindverse Computing LLC, Lynnwood, WA 98087, USA*

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Abstract

A complete authored walkthrough of rouché's theorem, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

CA-04 / Complex Analysis / Report v1

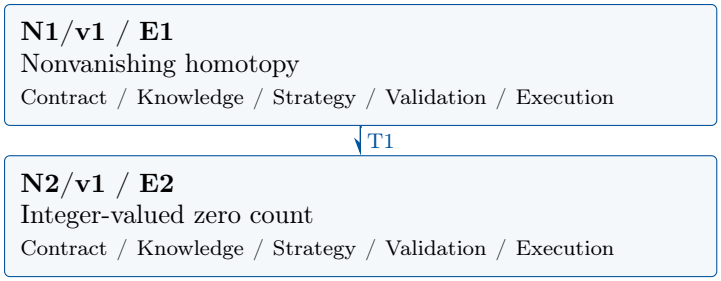
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *For a bounded Jordan domain with piecewise smooth boundary, let f, g be holomorphic near its closure and satisfy $|g| < |f|$ on the boundary. Prove f and $f + g$ have the same number of interior zeros, counted with multiplicity.*

A. Foundations and permitted imports

Import the argument principle: for holomorphic h near a bounded piecewise smooth Jordan domain closure and no boundary zero, $(2\pi i)^{-1} \oint h'/h$ is the number of interior zeros counted with multiplicity. It follows from the residue theorem by locally factoring each zero $h = (z - a)^m$ u with $u(a) \neq 0$. The domain has positive boundary orientation.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1: N1/E1/v1 to N2. A uniform boundary lower bound $\delta > 0$ for all h_t.

Receiving activity. N2: Integer-valued zero count

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Boundary lower bound; argument principle; prior exports E1/v1

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Nonvanishing homotopy

Object and role. Keep a strict lower bound for the entire deformation.

Inputs. $|g| < |f|$ on the compact boundary; foundational imports stated in §1

C — Contract

Own E1/v1: A uniform boundary lower bound $\delta > 0$ for all h_t.

K — Knowledge

Retain $|g| < |f|$ on the compact boundary; foundational imports stated in §1; keep their assumptions and versions attached.

S — Strategy

Keep a strict lower bound for the entire deformation.

V — Validation

Strict inequality is used on every boundary point.

E — Execution

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N1.1 Set $h_t = f + t\ g$ for $0 \le t \le 1$.

N1.2 On the boundary $|h_t| \ge |f| - t|g| \ge |f| - |g|$.

N1.3 The continuous positive difference has a positive minimum δ on the compact boundary.

N1.4 Thus every h_t is boundary-nonzero uniformly in t; no h_t is identically zero.

Export E1/v1. A uniform boundary lower bound $\delta > 0$ for all h_t.

Acceptance gate. Strict inequality is used on every boundary point. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N1: Strict inequality is used on every boundary point. Recheck dependent nodes N2.

B. N2/v1: Integer-valued zero count

Object and role. Use continuity in t and integrality.

Inputs. Boundary lower bound; argument principle; prior exports E1/v1

C — Contract

Own E2/v1: The number of zeros stays constant along the homotopy.

K — Knowledge

Retain Boundary lower bound; argument principle; prior exports E1/v1; keep their assumptions and versions attached.

S — Strategy

Use continuity in t and integrality.

V — Validation

A zero crossing the boundary would invalidate the invariant.

E — Execution

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N2.1 Define $N(t) = (2\pi i)^{-1} \oint (f' + t\ g')/(f + t\ \mathrm{g})\ \mathrm{dz}$.

N2.2 Numerator and denominator vary continuously on the compact contour and the denominator is bounded away from 0; hence N is continuous in t.

N2.3 By the argument principle each $N(t)$ is a nonnegative integer.

N2.4 A continuous map from the connected interval[0,1] into the integers is constant.

N2.5 Therefore $N(0) = N(1)$, proving equal zero counts for f and f+g.

Export E2/v1. The number of zeros stays constant along the homotopy.

Acceptance gate. A zero crossing the boundary would invalidate the invariant. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N2: A zero crossing the boundary would invalidate the invariant. Withhold this terminal result until its gate passes.

IV. CONTINUOUS MATHEMATICAL PROOF

Read these steps in order as one proof; the diagram is optional. Numbered activities identify the specific constructions and deductions owned by each node.

A. §4.N1 — Nonvanishing homotopy

1. Set $h_t = f + t$ g for $0 \le t \le 1$.
2. On the boundary $|h_t| \ge |f| - t|g| \ge |f| - |g|$.
3. The continuous positive difference has a positive minimum δ on the compact boundary.
4. Thus every h_t is boundary-nonzero uniformly in t; no h_t is identically zero.

B. §4.N2 — Integer-valued zero count

1. Define $N(t) = (2\pi i)^{-1} \oint (f' + t\ g')/(f + t\ \mathrm{g})\ \mathrm{dz}$.
2. Numerator and denominator vary continuously on the compact contour and the denominator is bounded away from 0; hence N is continuous in t.
3. By the argument principle each $N(t)$ is a nonnegative integer.
4. A continuous map from the connected interval[0,1] into the integers is constant.
5. Therefore $N(0) = N(1)$, proving equal zero counts for f and f+g.

□

V. WORKED ORCHESTRATION

On$|z| = 2$ compare $f = z^3$ and $g = z + 1$. Then $|g| \le 3 < 8 = |f|$. Thus $z^3 + z + 1$ has exactly three zeros *inside*$|z| < 2$, counted with multiplicity, although no individual root is located.

A. Boundary and hypothesis audit

With $f(z) = z$ and $g(z) = -1$ on the unit circle, only $|g| \le |f|$ holds. The endpoint $z - 1$ has a boundary zero. Validation rejects the missing strict margin; the zero-count homotopy is not admissible.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Failure and correction

This is an authored teaching trace, not a recorded agent run.

1. **Contract → Knowledge:** register the §1 theorem, imports, and requested exports before beginning the proof.
2. **Strategy → Execution:** choose the bridges in §3 and carry out the numbered activities in §4; transmit only established E-exports along the §2 DAG.
3. **Validation → responsible node:** test the specific boundary/failure in §5 and the node's individual audit. A failing candidate is withheld; downstream nodes may not use it.
4. **Knowledge + Strategy:** identify whether the error is an omitted condition, an invalid inference, or an attempted extension beyond the theorem. Restore the exact contract or explicitly state a different one.
5. **Execution:** redo the affected mathematical activity using the repaired condition or inference. Recheck consuming nodes, rather than feeding their conclusions back as premises.
6. **Contract:** publish only the supported export with its assumptions and proof location. A hypothetical revised export is v2; the written proof here is the v1 exposition, not a claim that this repair was executed.

B. Hypothesis boundary

With $f(z) = z$ and $g(z) = -1$ on the unit circle, only $|g| \leq |f|$ holds. The endpoint $z - 1$ has a boundary zero. Validation rejects the missing strict margin; the zero-count homotopy is not admissible.

C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading.

- Harvard complex analysis: Cauchy formula, argument principle, and Riemann mapping lectures.